Parvez Ali

MSc Candidate | Artificial Intelligence & Data Science | Medical Imaging Researcher

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RESEARCH INTERESTS

Medical image analysis retinal OCT segmentation and disease characterization longitudinal deep learning for treatment outcome prediction efficient 3D CNN architectures for volumetric data clinical decision support systems

RESEARCH PROFILE

MSc student specializing in AI-driven medical imaging, with 12+ months of applied research at DeepEye GmbH (Munich) on OCT-based analysis of neovascular AMD (nAMD). My thesis investigates deep learning models for predicting anti-VEGF treatment requirements from longitudinal OCT volumes, combining volumetric computer vision, clinical data understanding, and temporal modelling. I have hands-on experience designing image preprocessing pipelines, building 3D CNN training frameworks, and developing B-scan importance-sampling strategies for high-dimensional retinal data. I am seeking a PhD position at the intersection of AI and medical imaging to develop clinically translatable tools.

MSC THESIS

Prediction of Anti-VEGF Injections in nAMD Using Longitudinal OCTs 01 Dec 2025 – 30 Apr 2026
DeepEye GmbH / Deggendorf Institute of Technology – Munich, Germany

- Developing deep learning models to predict anti-VEGF injection requirements for nAMD patients from time-series OCT volumes, capturing longitudinal disease progression.
- Combining temporal modelling with volumetric image features to capture disease progression patterns across follow-up visits.

RESEARCH & INDUSTRY EXPERIENCE

Working Student – Computer Vision & Medical AI 01 Dec 2025 – Present
DeepEye GmbH – Munich, Germany

- Developing a B-scan importance-sampling algorithm to identify diagnostically relevant slices from large OCT volumes (500+ scans per eye), reducing pipeline inference time while preserving clinical accuracy.
- Conducting MSc thesis research on predicting anti-VEGF injection requirements in nAMD patients from longitudinal OCT data using deep learning.

Data Science Intern – Medical Imaging 15 May 2025 – 15 Nov 2025
DeepEye GmbH – Munich, Germany

- Performed exploratory data analysis and demographic distribution analysis on clinical tabular datasets using Plotly; built interactive stakeholder reports in Looker Studio.
- Researched and benchmarked OCT image preprocessing techniques (CLAHE, gamma correction, histogram equalization) to improve input quality for downstream segmentation models.
- Designed and evaluated B-scan pooling/sampling strategies to reduce 500+ scans to 20–40 representative slices; validated effects via 3D CNN training experiments on nAMD classification.
- Built an internal React + Flask dashboard to visualize product TPS (throughput) usage metrics.

Subject Specialist – Computer Science Aug 2023 – Mar 2024
IBA Community Colleges – Pakistan

- Taught Web Development (MERN stack) and Object-Oriented Programming (Java) to BSc Computer Science students.
- Developed self-paced tutorials and fostered collaborative learning environments.

EDUCATION

M.Sc. Artificial Intelligence and Data Science Mar 2024 – Present
Deggendorf Institute of Technology – Deggendorf, Bavaria

- GPA: 1.7
- Thesis: Prediction of Anti-VEGF Injections in nAMD Using Longitudinal OCTs (in collaboration with DeepEye GmbH, Munich)

B.E. Electrical Engineering (Computer Systems) Aug 2018 – Aug 2022
Sukkur IBA University – Sukkur, Sindh, Pakistan

- Cumulative GPA: 3.46 / 4.00
- Thesis: Voice Authentication Based on Deep Learning for Sindhi Language

RESEARCH PROJECTS

Satellite Imagery Road Segmentation

- Trained a U-Net model on the Massachusetts Roads Dataset (1,000+ images); 87% test accuracy. Applied to urban planning and infrastructure analysis.

Genomic Feature Analysis & Species Classification

- Bioinformatics pipeline for GC content, codon usage, and neutrality plot analysis across 3 species; Random Forest classifier for genomic feature-based species identification.

LLM Fine-Tuning: Extractive QA & Text Summarisation

- Fine-tuned BERT and DistilBERT (distilbert-base-cased-distilled-squad) on the SQuAD dataset for extractive question answering, model correctly identified answer spans from unseen contextual passages.
- Fine-tuned T5 for abstractive text summarization using a custom prompt-prefix approach (summarize:), demonstrating seq2seq training and tokenization strategies.
- Analyzed architectural differences between encoder-only (BERT) and encoder-decoder (T5) models, evaluating suitability for different NLP task types.

TECHNICAL SKILLS

Medical Imaging	OCT analysis, nAMD, retinal segmentation, CLAHE, gamma correction, volumetric 3D processing
Deep Learning	PyTorch, TensorFlow, CNN, 3D CNN, U-Net, RNN/LSTM, Transformer architectures
Computer Vision	Image segmentation, preprocessing pipelines, HuggingFace, Scikit-Learn
ML & Data	Pandas, Plotly, SQL, Looker Studio, Power BI, Apache Spark, HDF5
LLM & NLP	Fine-tuning (BERT, T5, DistilBERT), SQuAD QA, HuggingFace Transformers, RAG, LangChain, LangGraph, Llama 3.2, Agentic AI
Dev & Tools	Python, R, C/C++, React, Flask, Git, AWS, Linux
Languages	English (Fluent) | German (Intermediate) | Urdu (Native)

CERTIFICATIONS

- Event-Driven Agentic Document Workflows – DeepLearning.AI (Apr 2025)
- Open-Source Models with Hugging Face – DeepLearning.AI (Apr 2025)
- Sequences, Time Series and Prediction – Coursera (Aug 2023)